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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BCA (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER Discipline Specific Elective (DSE) - MG4DSEBCA200 - DATA VISUALIZATION (2024 ADMISSION ONWARDS)

Duration: 1 Hours **Maximum Marks: 35**

Scheme for Valuation

Part A - Very Short Answer Questions

Answer any 5 questions. Each question carries 3 marks

1. Explain why visual representation is more effective than textual reports in decision contexts.

Visual representation is more effective than textual reports in decision contexts because it leverages the human brain's natural ability to process visual information rapidly, often identifying patterns, trends, and anomalies at a glance that would take hours to uncover in raw data. Visual stories make information easier to understand and analyse and communicate. visuals reduce cognitive load. Instead of reading through dense paragraphs or tables, a chart or graph summarizes large amounts of data at a glance. This allows decision-makers to grasp key insights quickly without mentally stitching together scattered details.

2. Explain why selecting an appropriate visual display is important in data storytelling.

Selecting an appropriate visual display is important in data storytelling because it transforms complex, raw data into a clear, engaging, and accurate narrative that drives action. The right visual reduces cognitive load, allowing the audience to grasp insights in seconds rather than wasting time interpreting confusing or misleading graphs

3. Identify the two main types of numerical data? Explain briefly.

Numerical data is used to represent the height, weight, length and other things of the data. Two types of numerical data are

1. Discrete data – This is a type of data in statistics that only uses discrete values or single values . eg: Number of students in a class, Number of family members etc
2. Continuous data – it is a type data that represent the data in continuous range eg: weight of each student in a class, salary of workers in factory etc

4. State any three uses of the data visualization tool.

Data visualization tools are used to turn raw data into clear, actionable insights across many fields.

- **Used in business intelligence and decision-making**- Tools like Tableau and Microsoft Power BI help organizations track performance through dashboards, revealing trends in sales, customer behaviour, and operations
- **Used for data exploration and analysis** - Instead of scanning spread sheets, analysts use visuals to uncover hidden patterns, correlations, and anomalies
- **In communication and reporting**, visualizations make findings easier to present to stakeholders
- **Real-time monitoring**. - Many industries rely on live dashboards to track systems, such as website traffic, financial markets, or supply chains. Visual alerts and trend lines allow quick responses to changes or issues.
- **Public policy and governance**- visualizations help communicate data like population trends, economic indicators, or health statistics to the public.
- Used in marketing and customer analytics

5. Discuss three examples of geographical data used in real life.

Geographical data refers to data that represents the location shape and characteristics of features on earth's surface

- **Navigation and route planning**
  Apps like Google Maps use geographical data such as road networks, traffic conditions, and user location to suggest the fastest routes and estimate travel time.
- **Weather forecasting**
  Meteorological departments use geographic data like latitude, altitude, and satellite imagery to predict weather patterns
- **Urban planning and smart cities**
  City planners use geographical data to design roads, public transport, drainage systems, and zoning areas. Population density maps and land-use maps help decide where to build schools, hospitals, or parks.
- **Disaster management**
  During floods, earthquakes, or landslides, geographic data helps identify high-risk zones and plan evacuation routes.
- **Agriculture**
  Farmers and agricultural agencies use data on soil type, rainfall, and land elevation to decide which crops to grow and when to plant or harvest
- **Transportation and logistics**
  Delivery services use geographic data for route optimization, tracking shipments, and reducing fuel costs. GPS systems help companies manage fleets efficiently.
  Answer any three give marks

6. State any two importance of Data Visualization in Agricultural Research Application

Data visualization plays a central role in agricultural research because it turns complex, multi-layered farm data into insights that researchers and farmers can actually use.

- Understanding patterns in crops and environment
- Visualization dashboards allow researchers and policymakers to compare crop performance across regions, seasons, or farming techniques.
- Efficient use of geographic data
- Visual analytics can reveal anomalies such as sudden drops in crop health, pest infestations, or disease spread.
- Enhancing communication and collaboration
- Supporting sustainable farming practices
  Answer any two with description give marks

7. Identify scenarios where Python is preferred over BI tools.

Python is preferred over BI tools because it offers flexibility and depth that many BI platforms can't match. Some of the reasons are

- Unlimited customization and flexibility
- Advanced data processing
- Integration with machine learning
- Large data sets
- Reproducibility and version control
- Cost and open source
- Automation and scalability

Part B Short Answer Questions

Answer any 4 questions. Each Question carries 5 marks

8. Analyse how misleading the audience and perpetuating bias can impact public opinion?

If the student writes about how to know your audience and purpose give marks

9. Analyse the impact of biased data collection on business insights derived from visualization.

If the student writes anything related to this question give marks proportional to other answers

10. Explain the importance of using colour sparingly and consistently in data visualization with suitable examples.

Explanation of colour sparingly with example - 2.5 marks

Explanation of Colour Consistency with example - 2.5 marks

11. Discuss the importance of transformation and modelling in the analysis phase.

Explanation of data transformation- 2.5 marks

Explanation of data modelling – 2.5 marks

12. Describe the importance of data modelling in enhancing the performance of visualization tools.

- Ensure data quality & Consistency
- Define relationships and context
- Improve performance and efficiency
- Enables self-service Analytics
- Support Scalability and adaptability
  If the student writes all these points with explanation give full marks
  If the student writes points only give 2 marks

13. What are the major challenges when dealing with complex data in visualization?

Challenges when dealing with complex data in data visualization are

- Handling large data sets
- Data quality and inconsistency
  Challenges in data visualization with above mentioned points give full marks
  If the student writes challenges only give 3 marks

Scheme prepared by

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